Presentation Script

Title slide *(∼10 sec)*

> Hi, I'm **Gurpreet Singh**, this is **Md Rafi**. Our project predicts whether a chemical reaction will succeed or fail, using machine learning combined through an ensemble framework called **Combinatorial Fusion Analysis**.

Background note (only if asked): Our model pool has 51 learners: 41 fingerprint-based regressors — sklearn-style models like gradient-boosted trees, kernel methods, MLPs, linear regressors — that predict continuous yield from the molecular fingerprint, then we threshold at 50% to get a binary label. Plus 5 native classifiers trained directly on binary labels, plus the deep graph-based chemistry model Chemprop.

Slide 1 — The Problem *(∼60 sec)*

> The reactions we study are **palladium-catalyzed Buchwald-Hartwig aminations**, used heavily in pharmaceutical synthesis to form carbon-nitrogen bonds — they couple an **aryl halide** with an **amine** to make a new C-N bond.

Definitions (deliver inline if needed):

- An **aryl halide** is an aromatic ring (think benzene-like, six carbons in a hexagon) with a halogen atom — Cl, Br, I, or a triflate group — attached. The carbon-halogen bond is what gets broken in the reaction.
- A **primary amine** has nitrogen bonded to one carbon and two hydrogens (R-NH_2). A **secondary amine** has nitrogen bonded to two carbons and one hydrogen (R-NH-R'). The N-H bond is what reacts to form the new C-N bond.

> Each reaction has **six ingredients** we encode as SMILES strings — a textual representation of molecular structure — plus a reaction temperature. The six are: the **aryl halide**, the **amine**, the **palladium catalyst** (with its phosphine ligand), the **base** (e.g. cesium carbonate), the **solvent** (e.g. toluene or dioxane), and an optional **additive**.

> We define **success as yield $\geq$ 50%**. This is a working convention from process chemistry, not a hard scientific threshold — it's the "preparatively useful" line below which side reactions dominate and the synthesis isn't worth scaling up.

Reference (deliver only if pressed): There's no single canonical paper for the 50% line; it's a convention. Closely related: Carey et al. (2006), *Organic & Biomolecular Chemistry*, "Analysis of the reactions used for the preparation of drug candidate molecules" — a survey of pharma reaction yields. Anderson, N.G. (2012) *Practical Process Research and Development* (textbook) discusses yield thresholds in process development.

> The use case is bench-chemistry triage: predict which reactions are worth running before committing the lab time.

Slide 2 — Why classification, not regression? *(∼55 sec)*

> The first version of this project predicted exact yield as a continuous number. On a random train/test split it worked great — R-squared 0.82, competitive with published benchmarks.

> But on **held-out scaffolds** every model collapsed below R-squared zero — meaning none of them did better than just predicting the mean.

Definition (deliver inline): A **scaffold** is the core ring framework of a molecule with all the hanging substituents stripped off. For example, the molecule "4-bromobenzonitrile" is a benzene ring (the scaffold) with a bromine and a cyano group attached. Two molecules sharing the same benzene scaffold belong to the same scaffold group even if their substituents are completely different. "Held-out" means we partitioned the data so the test set has scaffolds the model never saw at training. The formal definition we use is the **Bemis-Murcko scaffold** (Bemis & Murcko, 1996, *J. Med. Chem.* — "The properties of known drugs. 1. Molecular frameworks").

> The pivot to binary classification is more honest because chemists making a go/no-go decision **might not** need exact yield precision — they need to decide whether the reaction is worth attempting.

Slide 3 — Dataset *(∼75 sec)*

> 5,465 reactions from the **Open Reaction Database, or ORD** — an open-science chemistry repository where labs submit experimental reaction data in a standard format.

Why we did feature construction (deliver inline): Sklearn-style models — gradient-boosted trees, linear models, MLPs — work on fixed-length numeric vectors, not on molecular graphs. So we need to convert each molecule into a numeric fingerprint that encodes its structure. The choice of featurization determines what information the model has access to.

> For feature construction we use **Morgan fingerprints**: 2,048-bit binary vectors that record which chemical substructures appear in a 2-atom radius around each atom. We compute one per reaction component — six total — and concatenate **DRFP**, a reaction-level fingerprint that encodes the difference between reactants and products as a single vector. About 14,000 features per reaction.

> Two splits. **Random** is the standard 70/15/15 shuffle — in-distribution. **Substrate-level 5-fold cross-validation** is our out-of-distribution test.

Definitions (deliver inline):

- A **substrate** in chemistry is a molecule that participates in a reaction — typically the molecule that gets modified. Here, the aryl halide is the substrate (the partner that gets aminated).
- **Substrate-level 5-fold CV vs regular k-fold**: regular k-fold just shuffles all rows and partitions them into 5 disjoint groups — the same substrate can appear with different reactions in both train and test, so the model has effectively seen the compound during training. *Substrate-level* k-fold groups by the full aryl-halide structure first (we have 365 unique substrates), then bin-packs whole groups into 5 folds. So if a substrate is in the test fold, *all* of its reactions are in test and *none* are in train — the model has truly never seen that compound. Both protocols give 5 train/test splits. **The difference is just how we group rows.**
- **OOD = out-of-distribution**, a general ML concept (popularized in deep learning by Hendrycks & Gimpel, 2017, *ICLR*). It means the test data is structurally different from the training data, so the model

has to extrapolate rather than interpolate. ORD is just the dataset name; OOD is the test regime — they're unrelated.

> The substrate-level methodology is standard in chemistry ML — see Yang et al. 2019, the Chemprop paper, and Wu et al. 2018, MoleculeNet.

Slide 4 — Approach *(~30 sec)*

> SMILES and temperature go through featurization, hit a 51-model pool, and CFA selects the best ensemble. CFA is from Lin et al. 2007 — it scores models on individual accuracy AND **cognitive diversity**, which measures how differently two models distribute their prediction scores. CFA explores ensemble sizes 2 through 5 across 6 weighting methods.

Slide 5 — Random-split results *(~45 sec)*

> Top 15 models on the **random split** — meaning shuffled 70/15/15 — with **95% bootstrap confidence intervals**. A bootstrap CI works like this: we resample the test set 1000 times with replacement, recompute accuracy each time, and the 2.5th and 97.5th percentile give the interval. Translation: if we drew a slightly different test set, the accuracy would land between these bars 95% of the time.

Reference (deliver only if asked): The bootstrap method is from Efron (1979), *Annals of Statistics*, "Bootstrap Methods: Another Look at the Jackknife". A more accessible treatment is Efron & Tibshirani (1993), *An Introduction to the Bootstrap* (CRC Press textbook). It's now the standard non-parametric way to put error bars on any test-set statistic.

> Top models cluster between 0.91 and 0.93. LightGBM and the native LightGBM (Light Gradient Boosting Machine) classifier are statistically tied — their CIs overlap completely.

Slide 6 — Substrate-level OOD *(~45 sec)*

> The harder test, 5-fold CV by substrate. Box plots show per-fold accuracy across the 5 folds. **Native_XGBClf**, the XGBoost classifier, is the best individual at 0.824 ± 0.047 accuracy. **Native_LGBClf**, the LightGBM classifier, is just behind. The CFA ensemble lands at 0.804 with tighter spread.

> **Selection stability**: across the 5 folds, did CFA pick the same models? Yes — Native_LGBClf and Native_XGBClf chosen in 5 out of 5 folds. The choice is robust, not luck we got from choosing some seed.

Slide 7 — Does CFA earn its complexity? *(~50 sec)*

> We compared CFA against simpler ensembling: single-best model, equal-weight top-k averaging, and majority voting. Bars go from worst on the left to best on the right.

> "Random regression" and "random classification" mean those tasks evaluated **on the random split**. On random regression, CFA wins by a small but real margin — 1.2 R-squared points over equal-weight top-5. On random classification, CFA ties majority voting almost exactly. On the scaffold splits, the validation set has only 10 reactions, so CFA can't reliably choose between candidates.

> For an honest takeaway: CFA earns its complexity on tighter regression problems; for large-validation classification, simple averaging works just as well. We don't oversell it.

Slide 8 — Robustness to the success threshold *(∼30 sec)*

> The 50% success threshold is the chemistry literature's "preparatively useful" line — below 50%, side reactions consume too much material to scale up. To make sure our results aren't an artifact of that exact choice, we swept the threshold from 20% to 80% in 5% steps. The (MCC) Matthews correlation coefficient stays above 0.78 across the entire 50–75% range — the model is robust to where we draw the line.

Slide 9 — Summary *(∼55 sec)*

> What we built — three lines: a 51-model pool combined through CFA; two evaluation regimes — random in-distribution and substrate-level cross-validation for OOD; a full reliability stack — bootstrap intervals, multi-seed reproducibility, McNemar paired tests, and an ablation. **92% accuracy in-distribution, 80% on unseen substrates.**

> Honest limitations:

> First, the **Murcko-scaffold OOD test**. This is a stricter test than substrate-level CV: instead of holding out specific compounds, we hold out entire ring systems. So the model trains on benzene derivatives and pyridine derivatives, then has to predict on completely new ring systems like quinoline or pyrazole. On our dataset, accuracy drops to 59% because 88% of the data uses just two scaffolds — benzene and pyridine — so the train set is dominated by them and the test set is everything exotic. The validation set in this regime ends up with only 10 reactions, which is too few to tune any hyperparameter.

> Second, **ORD has reporter bias** — high-yield reactions get published more often than low-yield ones, so any classifier trained on the database inherits that imbalance. Our dataset is roughly 67/33 failure-to-success, but the real chemistry-space distribution is unknown.

> Thank you.

Backup slides ***(used only if asked)***

- **Multi-seed reliability** — Native_LGBClf reproducible to ±0.01 accuracy on random, ±0.06 on OOD.
- **Native vs thresholded regressor** — native classifiers tie tree families but reliably beat the linear family.
- **Rank-score curves** — CFA's signature plot. Each model normalized to [0,1] and sorted descending; dashed line is the oracle.
- **Paired statistical tests** — McNemar p-values; Native_Logistic reliably beats Ridge ($p = 0.036$).
- **Confusion matrix** — errors balanced across classes despite 67/33 imbalance.
- **Learning curves** — substrate-OOD accuracy is still rising at 100% of available data — more data would help OOD.